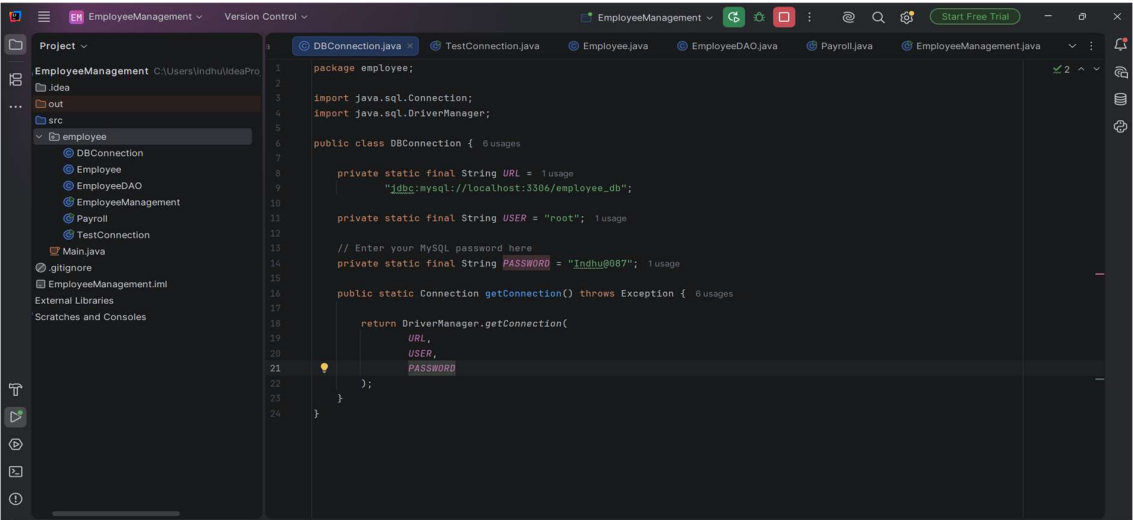
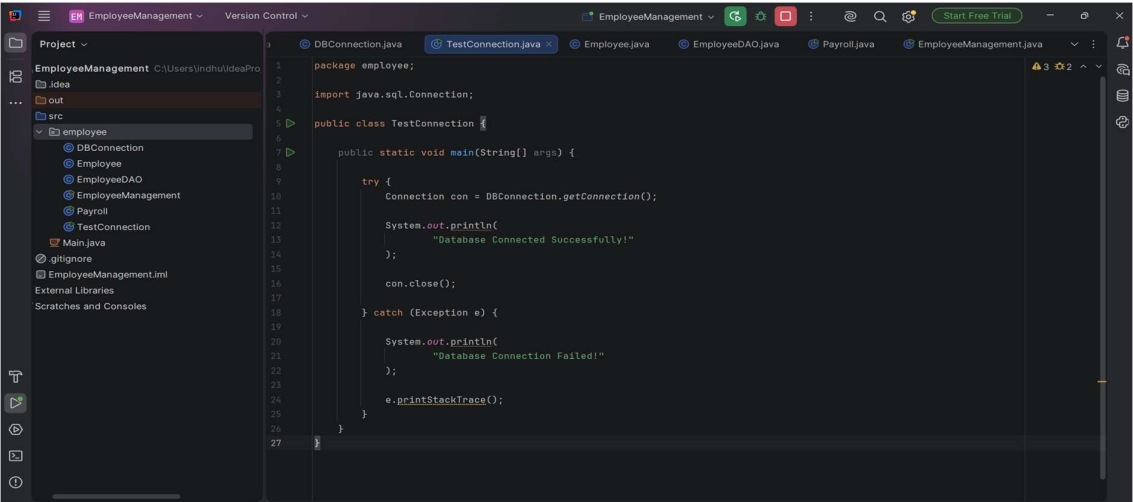


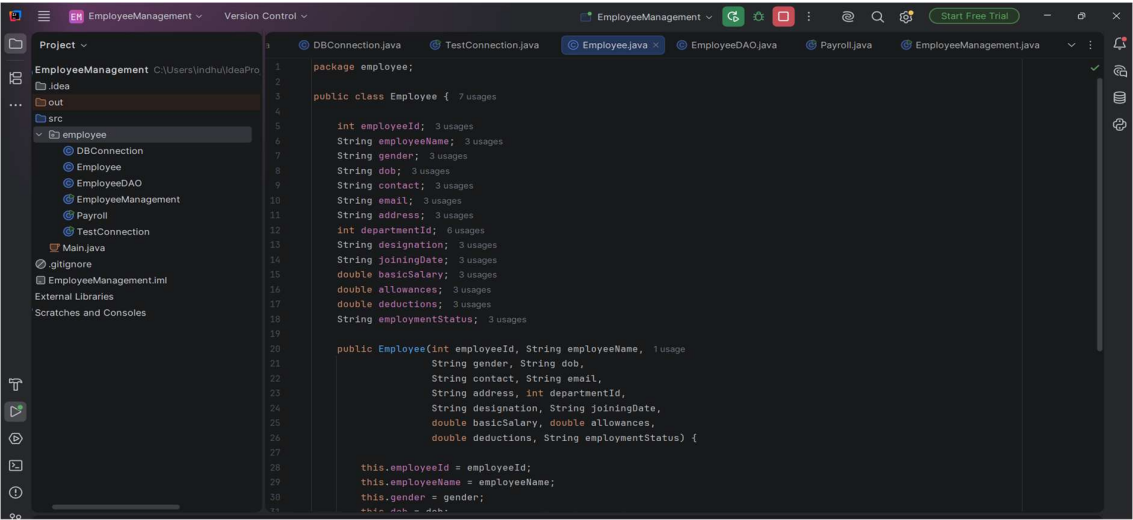
Employee Management and Payroll Application Implementation and Testing



```
1 package employee;
2
3 import java.sql.Connection;
4 import java.sql.DriverManager;
5
6 public class DBConnection { @ usages
7
8     private static final String URL = 1 usage
9         "jdbc:mysql://localhost:3306/employee_db";
10
11     private static final String USER = "root"; 1 usage
12
13     // Enter your MySQL password here
14     private static final String PASSWORD = "Indhu@087"; 1 usage
15
16     public static Connection getConnection() throws Exception { @ usages
17
18         return DriverManager.getConnection(
19             URL,
20             USER,
21             PASSWORD
22         );
23     }
24 }
```



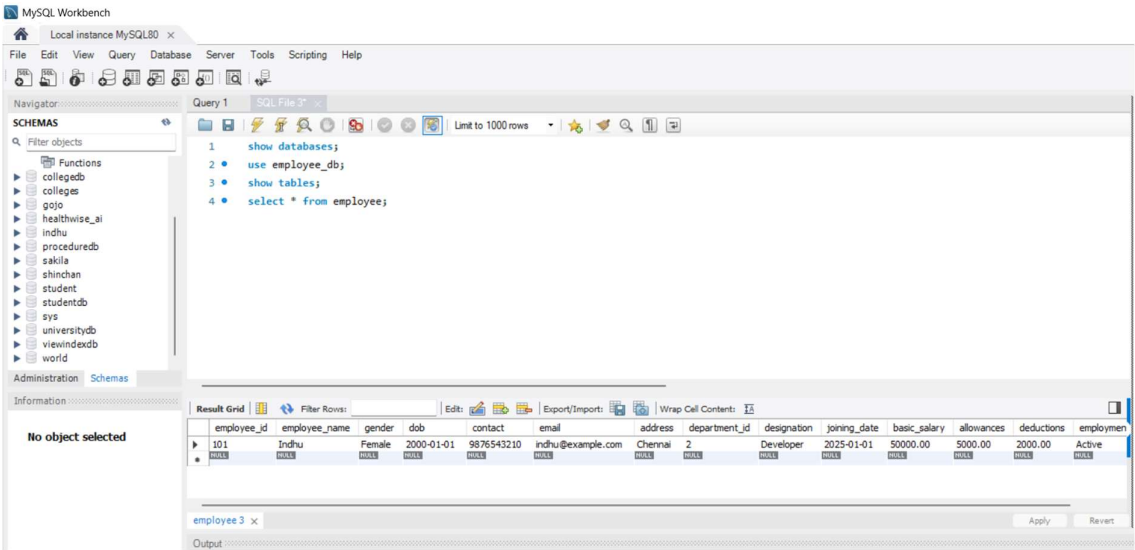
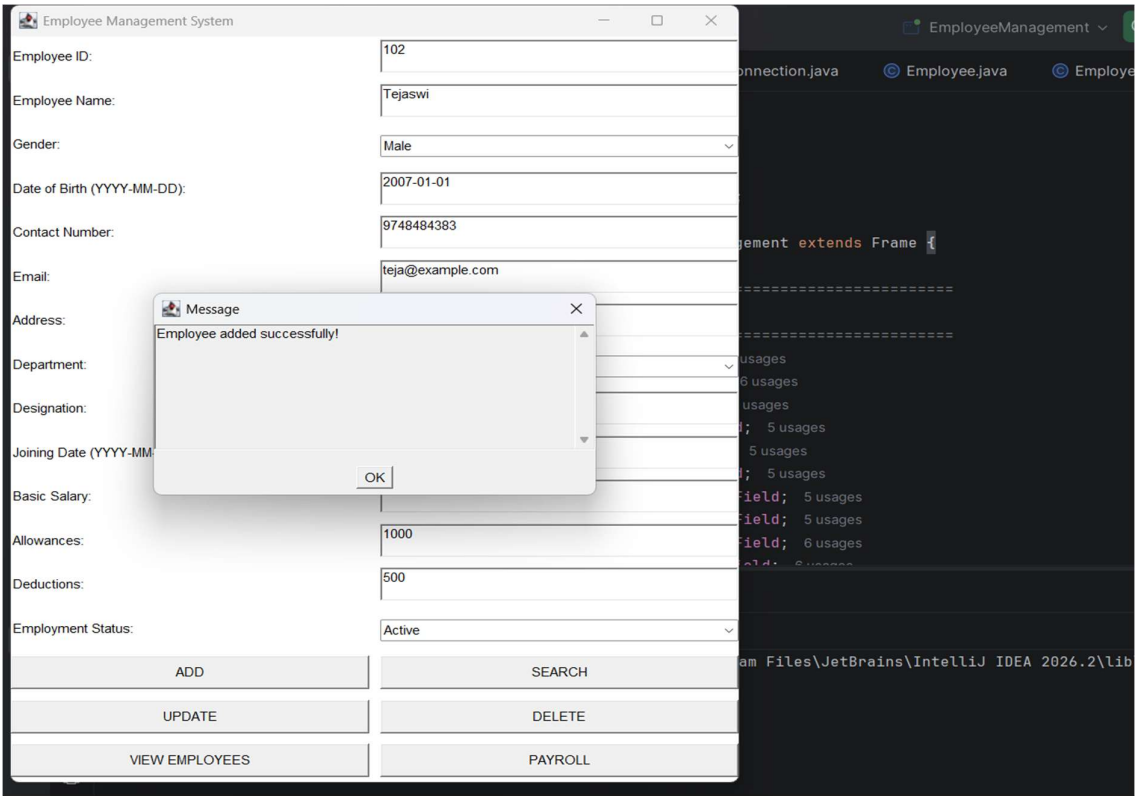
```
1 package employee;
2
3 import java.sql.Connection;
4
5 public class TestConnection {
6
7     public static void main(String[] args) {
8
9         try {
10             Connection con = DBConnection.getConnection();
11
12             System.out.println(
13                 "Database Connected Successfully!"
14             );
15
16             con.close();
17
18         } catch (Exception e) {
19
20             System.out.println(
21                 "Database Connection Failed!"
22             );
23
24             e.printStackTrace();
25         }
26     }
27 }
```

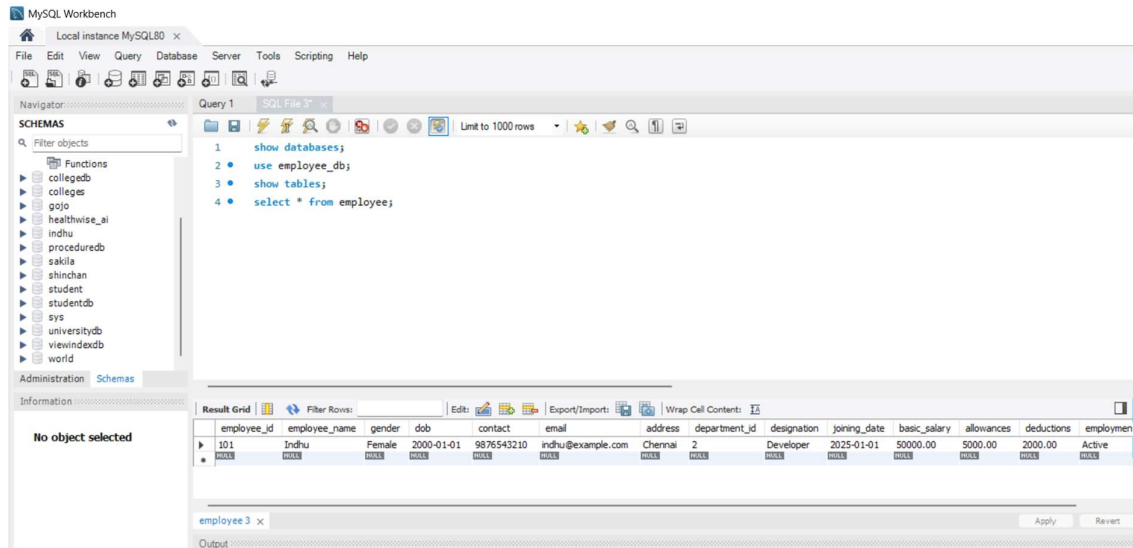


```
1 package employee;
2
3 public class Employee { 7 usages
4
5     int employeeId; 3 usages
6     String employeeName; 3 usages
7     String gender; 3 usages
8     String dob; 3 usages
9     String contact; 3 usages
10     String email; 3 usages
11     String address; 3 usages
12     int departmentId; 6 usages
13     String designation; 3 usages
14     String joiningDate; 3 usages
15     double basicSalary; 3 usages
16     double allowances; 3 usages
17     double deductions; 3 usages
18     String employmentStatus; 3 usages
19
20     public Employee(int employeeId, String employeeName, 1 usage
21         String gender, String dob,
22         String contact, String email,
23         String address, int departmentId,
24         String designation, String joiningDate,
25         double basicSalary, double allowances,
26         double deductions, String employmentStatus) {
27
28         this.employeeId = employeeId;
29         this.employeeName = employeeName;
30         this.gender = gender;
31     }
32 }
```


Employee Registration Success:

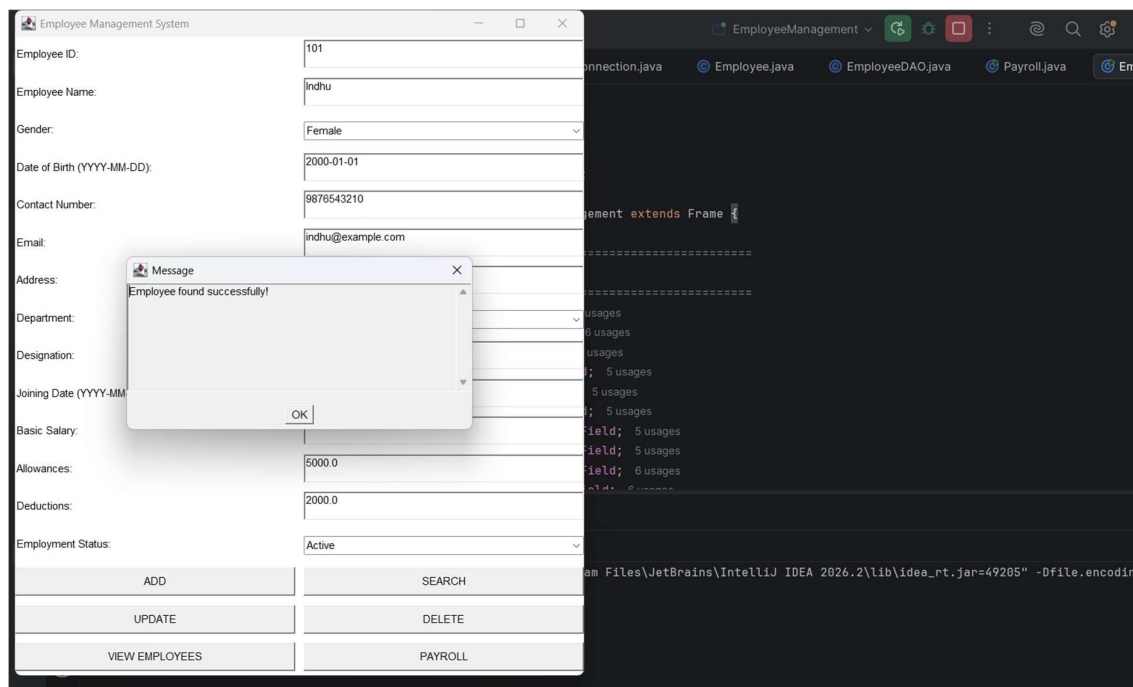
The employee details are successfully validated and inserted into the MySQL database using JDBC. A confirmation message is displayed to the user after successful registration.





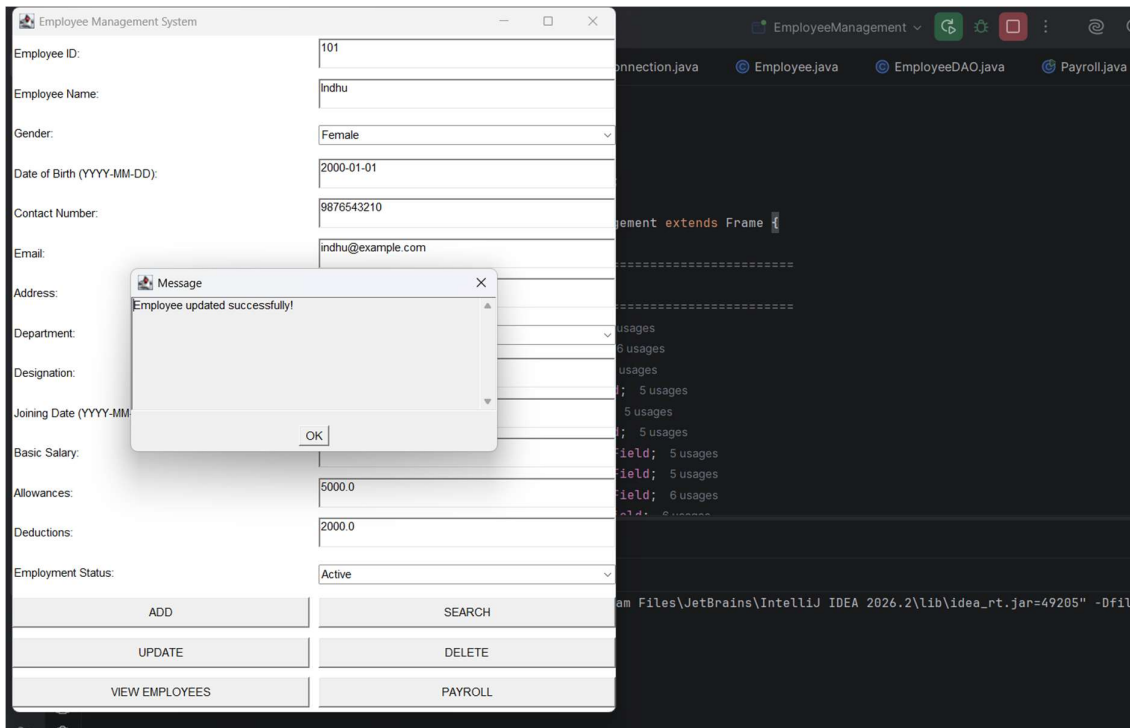
Employee Search:

This screen allows the HR staff to search for an employee using the Employee ID. The application retrieves the corresponding employee information from the MySQL database.



Employee Update:

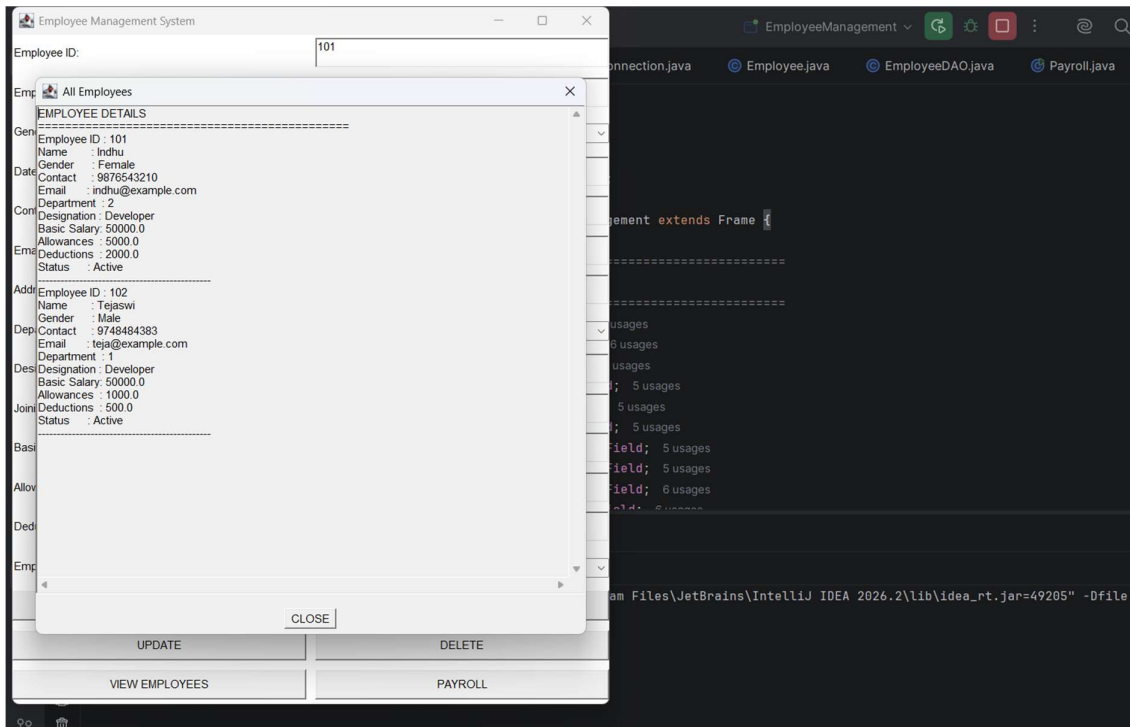
This screen is used to modify the existing employee information. The updated details are stored in the MySQL database using the JDBC UPDATE operation.



Employee Details:

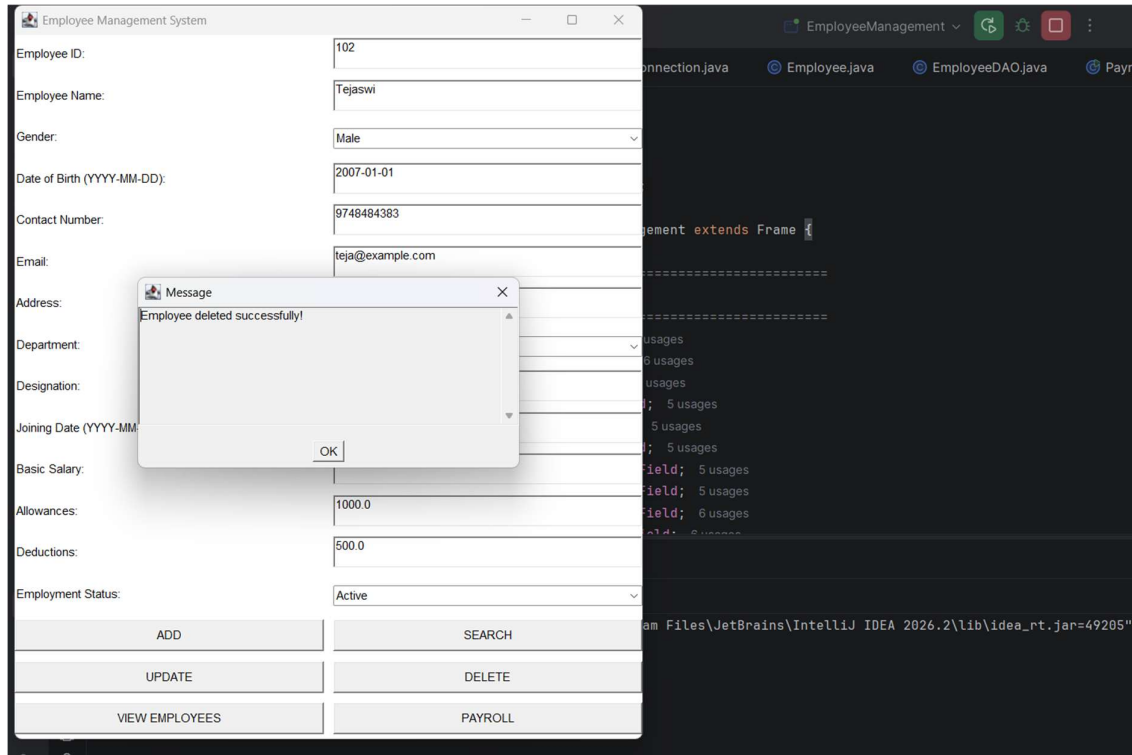
The selected employee's information is displayed through the AWT interface.

This demonstrates the retrieval of employee records from the database.



Employee Delete:

This operation allows the HR staff to remove an existing employee record from the database using the Employee ID.



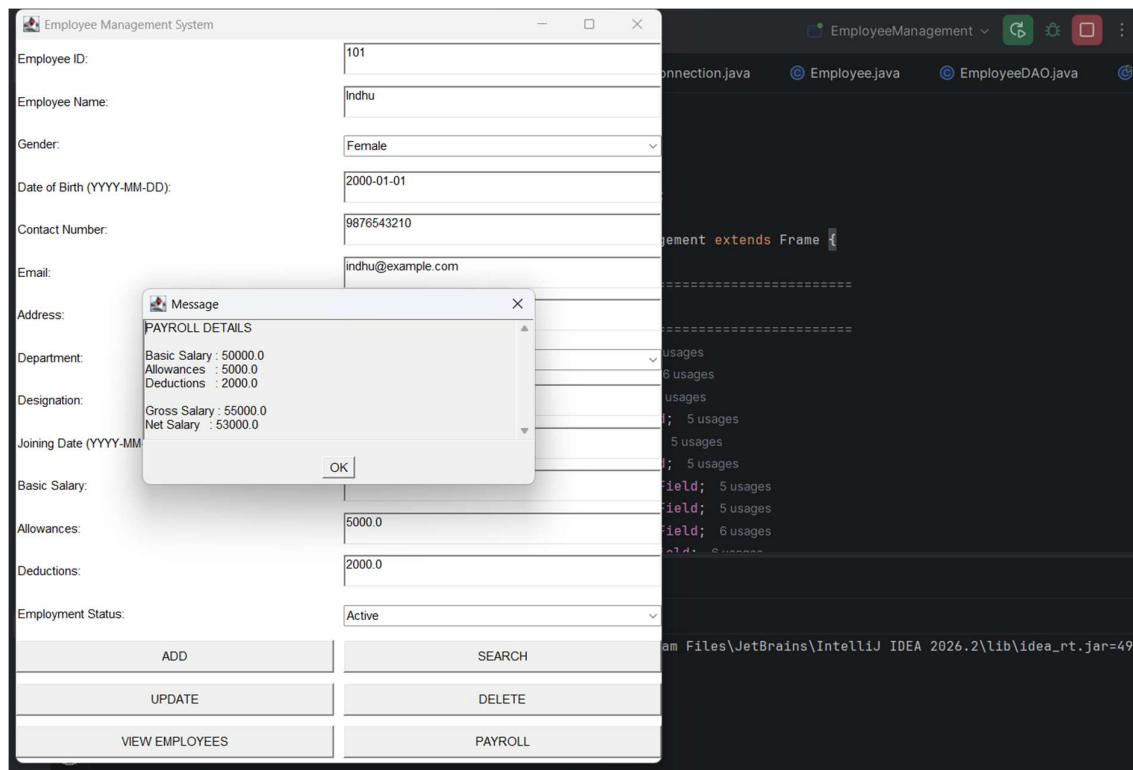
Payroll Calculation:

The payroll module calculates the employee's salary based on the given salary components.

Formula:

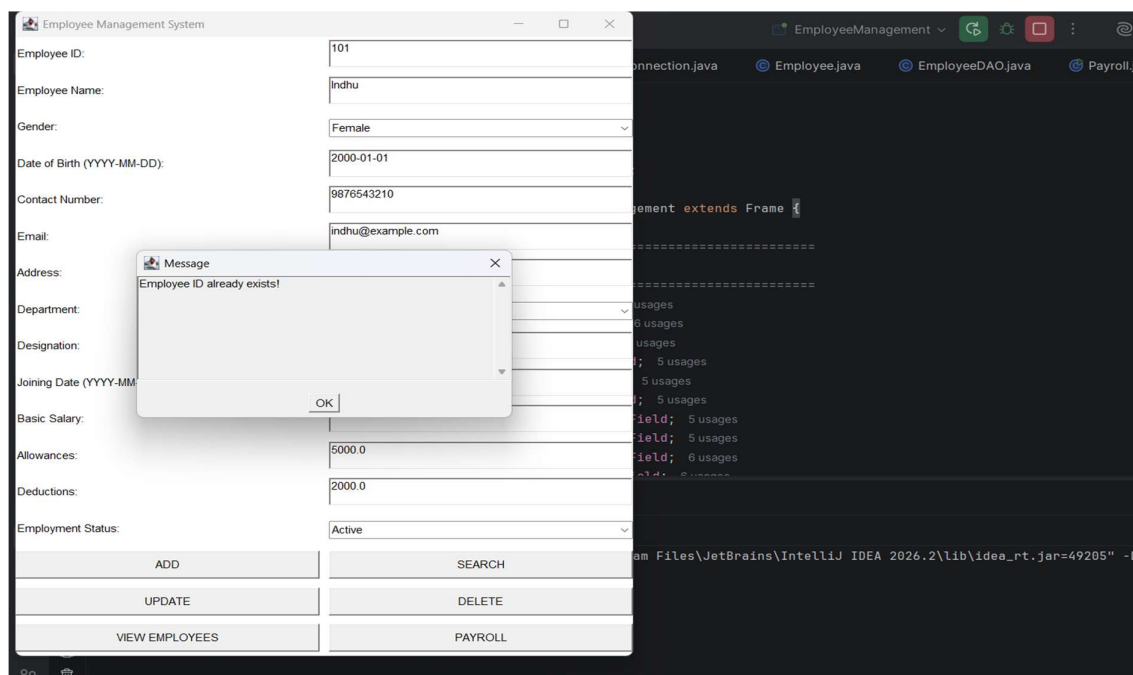
Gross Salary = Basic Salary + Allowances

Net Salary = Gross Salary – Deductions



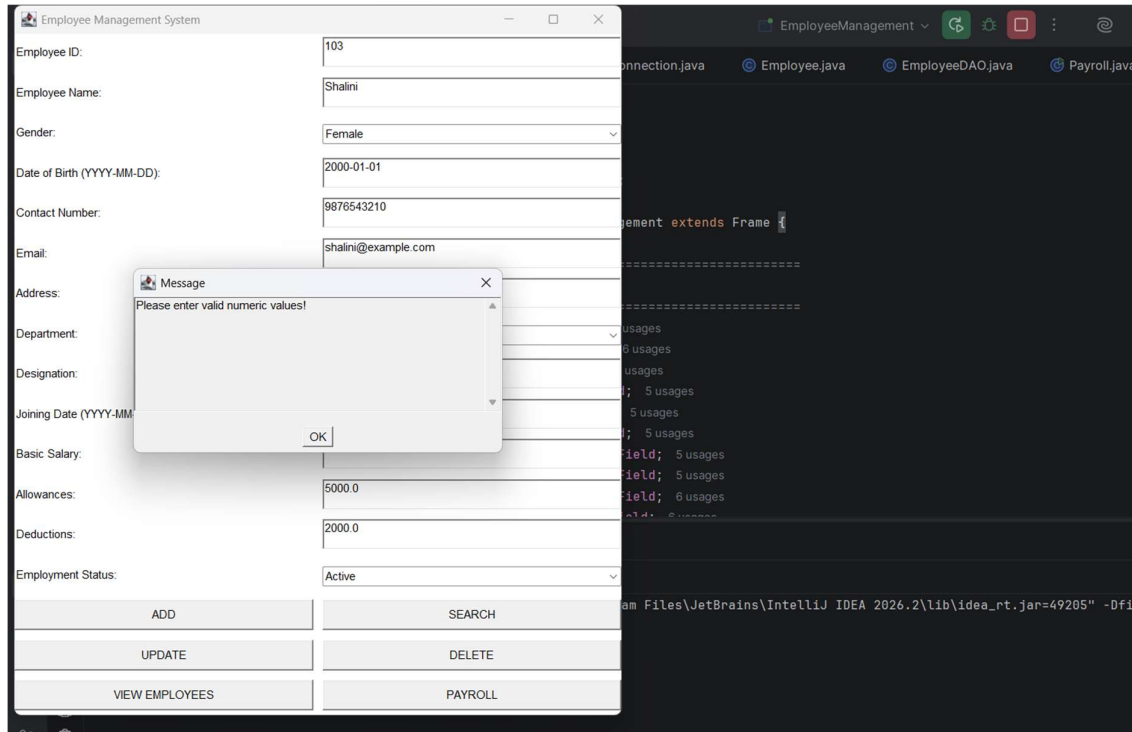
Duplicate Employee ID Validation:

The application prevents registration when an Employee ID that already exists in the database is entered. An appropriate error message is displayed to the user.



Invalid Input Validation:

The application validates user input and prevents invalid values from being stored. Appropriate error messages are displayed when incorrect data is entered.



Mandatory Field Validation:

The application checks required fields before performing database operations. If any mandatory field is empty, the user is notified and the operation is stopped.

